

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Show that	(L1/CO1)	6
a) $r(n) = \sum_{k=0}^{\infty} k \delta(n - k)$		
b) $r(n) = \sum_{k=-\infty}^{n-1} u(k) = \sum_{m=1}^{\infty} u(n - m)$		
B) Write any 6 properties of Discrete Time Fourier Transform.	(L1/CO2)	6
C) Find the energy of sequence	(L1/CO1)	6
$x(n) = \text{sinc} \frac{\omega_c n}{\pi}$		
Q.2 Solve Any Two of the following.		12
A) Find the Fourier transform of given signals. Also plot magnitude and phase.	(L2/CO2)	6
a) $x(n) = \delta(n)$		
b) $x(n) = a^n u(n)$		
B) a) Compute the 4-point DFT of $x(n) = (-1)^n \dots 0 \leq n \leq 3$ using matrix method.	(L3/CO1)	6
b) Perform linear convolution using circular convolution of given signals.		
$x(n) = \{2, 5, 0, 4\}, \quad h(n) = \{4, 1, 3\}$		
C) Given $x(n) = \{1, 0, -1, 0, 1, 0, -1, 0\}$. Find X(k) using DIT-FFT algorithm.	(L3/CO4)	6
Q. 3 Solve Any Two of the following.		12
A) Find the inverse z-transform using partial fraction method for given signal.	(L2/CO2)	6
Also plot the ROC with locations of poles that you calculated.		
$X(Z) = \frac{1}{(1-1.5Z^{-1}+0.5Z^{-2})}$		
if a) ROC: $ z > 1$		
b) ROC: $ z < 0.5$		

- B)** Use initial value theorem to find the initial value of the signals. **(L3/CO1)** **6**

a) $X(Z) = \frac{2+Z^{-1}}{(1-Z^{-1})(1+0.5 Z^{-1})}$

b) $X(Z) = \frac{1-3 Z^{-1}}{(1-0.1 Z^{-1})(1+0.6 Z^{-1})}$

- C)** Determine the Z-transform of given signal. Depict ROC and locations of poles and zeros in Z plane. **(L3/CO4)** **6**

$$x(n) = n\left(\frac{1}{2}\right)^{|n|}$$

Q.4 Solve Any Two of the following. **12**

- A)** Differentiate between Finite Impulse Response (FIR) filter and Infinite Impulse Response (IIR) filter. **(L2/CO2)** **6**

- B)** Transform the analog filter transfer function into digital filter $H(Z)$ using Impulse Invariant Method **(L3/CO1)** **6**

$$H_a(S) = \frac{0.5(S+4)}{(S+1)(S+2)}$$

- C)** For transfer function $H(S)$ find $H(Z)$ using Bilinear transformation method. Assume $T=1$. **(L2/CO2)** **6**

$$H(S) = \frac{1}{(S^2 + \sqrt{2}S + 1)}$$

Q. 5 Solve Any Two of the following. **12**

- A)** Explain the design method for a low pass Butterworth filter using a bilinear transformation. Given **(L3/CO2)** **6**

$$\begin{aligned} \delta_1 \leq |H(e^{j\omega})| \leq 1 & \quad 0 \leq |\omega| \leq \omega_p \\ |H(e^{j\omega})| \leq \delta_2 & \quad \omega_s \leq |\omega| \leq \pi \end{aligned}$$

- B)** Find the filter order N and cut-off frequency Ω_c for an IIR low pass Butterworth filter using Bilinear transformation. The filter specifications are as follows: **(L2/CO4)** **6**

$$\begin{aligned} \text{Passband: } 0.8 \leq |H(e^{j\omega})| \leq 1 & \quad |\omega| \leq 0.2\pi \\ \text{Stopband: } |H(e^{j\omega})| \leq 0.2 & \quad 0.6\pi \leq |\omega| \leq \pi \end{aligned}$$

Assume $T=1$ second.

- C)** Consider the following LTI system with system function and draw the direct form I and direct form II structure **(L2/CO2)** **6**

a) $H(Z) = 1 - \frac{1}{3}Z^{-1} + \frac{1}{6}Z^{-2} + Z^{-3}$

b) $H(Z) = \frac{1+2Z^{-1}-Z^{-2}}{1+Z^{-1}-Z^{-2}}$

***** End *****